

Minimum Subspace Trades and Relation Lattices in Three Binary Designs

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Abstract

We study the point–block relation modules of three fixed collections of four-dimensional subspaces of $\mathbb{F}_2^8$, each invariant under the same group of order 204. The minimum point trades are the six-block opposite-regulus trades. We determine their exact censuses, prove that they span the full rational relation module in each of the three configurations, and identify the complete degree-three layer of the associated incidence toric ideals. The minimum trades are locally saturated at 2, 3, 17, 137, 219 097, and 3 288 036 131. Exact complete-equation certificates cover every large component localization at each of the three selected good primes. Beyond these finite screens, two exact characteristic-zero minors from different row and column bases generate the unit ideal after 17 is inverted. Consequently the complete $\rho/\mathcal{B}_1$ equation module has structural rank 1312 over $\mathbb{Z}[\zeta_{17}, 1/(2 \cdot 3 \cdot 17)]$, so its residual component obstruction vanishes there. The corresponding global gates for the other five large maps and the global Smith groups remain open.

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Contents

1	Introduction	2
2	The locked triple and its common symmetry	4
2.1	Immutable identification of the three designs	4
2.2	The eight rational types	4
3	Rational relation modules	5
4	Minimum subspace trades	6
4.1	The opposite-regulus model	6
4.2	Exact census	7
5	Full rational generation	7
5.1	Component equation matrices	7
5.2	The exact rank certificate	8
6	The first toric layer	9
7	Integral relation lattices	10
7.1	The prime 2	10
7.2	The prime 3: the liftable kernel	11
7.3	The prime 17: a free cyclic kernel	12
7.4	What the small components exclude	12

8	Complete saturation at three selected good primes	13
8.1	The prime 137	13
8.2	The prime 219 097	14
8.3	The prime 3 288 036 131	15
9	A global localized theorem for the $\rho/\mathcal{B}_1$ component	15
9.1	Exact lift and determinant reconstruction	16
9.2	The determinantal-ideal certificate	17
10	Discussion, prior art, and open directions	18
10.1	The rational–integral contrast	18
10.2	Tomography and the P42 interface	18
10.3	Remaining global integral gates	19
10.4	Geometric and family questions	19
A	Exact certificate contracts	19
A.1	Characters and relation modules	19
A.2	Minimum-trade census	20
A.3	Rational component ranks	20
A.4	Local saturation	21
A.5	The 137-component	21
A.6	The 219 097-component	21
A.7	The 3 288 036 131-component	22
A.8	Global $\rho/\mathcal{B}_1$ structural rank	22
B	Reproducibility and source boundary	23

1 Introduction

Let $\mathcal{P}$ be the set of 255 points of $\text{PG}(7, 2)$. We fix three block collections $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, each consisting of 66 929 four-dimensional vector subspaces of $\mathbb{F}_2^8$. They arise as three constituents in the locked presentation studied in the companion construction paper; their construction lineage is described in [10]. The present paper starts from that finite triple and studies a different layer: relations among its point-incidence columns.

Write

$$A_i : \mathbb{Z}[\mathcal{B}_i] \longrightarrow \mathbb{Z}[\mathcal{P}], \quad B \longmapsto \sum_{x \in \text{PG}(B)} x.$$

Every A_i has rank 255. We compare the rational relation module $K_i = \ker(A_i \otimes \mathbb{Q})$ with the submodule generated by the smallest possible point trades. Those trades have three blocks in each leg and come from the two reguli in a quotient $\text{PG}(3, 2)$.

The three configurations have the same parameters and a common symmetry group

$$\Gamma \cong (C_{17} \rtimes C_4) \times C_3, \quad |\Gamma| = 204,$$

but their relation modules are not isomorphic. This is already visible in every rational irreducible type. Despite that separation, the same local six-block geometry generates all rational relations in all three cases.

Theorem 1.1 (Main theorem). *For the locked triple $(\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3)$, the following hold.*

- (i) *The rational $\mathbb{Q}\Gamma$ -modules K_1, K_2, K_3 are pairwise nonisomorphic, and their multiplicities differ in each of the eight rational irreducible types.*
- (ii) *The $\mathbb{Q}$ -span of the minimum opposite-regulus trades is K_i for every i .*
- (iii) *The associated incidence toric ideal has no nonzero element of degree below three. Its primitive cubics are precisely the minimum opposite-regulus trades; every such cubic is indispensable.*
- (iv) *If L_i is the integral lattice generated by the minimum trades and*

$$\Theta_i = \ker_{\mathbb{Z}}(A_i)/L_i,$$

then Θ_i is finite and has zero 2-, 3-, and 17-primary parts.

- (v) *The complete primary integral quotients vanish at all three selected good primes:*

$$\Theta_i[p^\infty] = 0 \quad (i = 1, 2, 3, \ p \in \{137, 219\,097, 3\,288\,036\,131\}).$$

- (vi) *Let $E_{\rho,1}^{\text{full}}$ be the complete $\rho/\mathcal{B}_1$ minimum-circuit equation matrix over $R = \mathbb{Z}[\zeta_{17}]$. Its order-1312 determinantal ideal satisfies*

$$I_{1312}(E_{\rho,1}^{\text{full}})R \left[\frac{1}{2 \cdot 3 \cdot 17} \right] = R \left[\frac{1}{2 \cdot 3 \cdot 17} \right].$$

Hence the complete equation module has its structural rank 1312 at every prime ideal away from the rational primes 2, 3, and 17, and the residual $\rho/\mathcal{B}_1$ component obstruction vanishes there.

The finite-prime assertions at 137, 219 097, and 3 288 036 131 are complete for the frozen triple. The last item is stronger but narrower in a different direction: it is global away from 2, 3, and 17, yet applies only to the complete $\rho/\mathcal{B}_1$ component. The corresponding global gates for the other five large maps remain open, so no support theorem or global Smith classification for the complete groups Θ_i is asserted. Likewise, our toric conclusion concerns the first nonzero graded layer; it does not assert that all higher-degree moves are generated by cubics.

The proof has three interfaces. Geometry identifies and counts the minimum circuits. Semisimple representation theory converts selected circuits into exact component-rank problems and proves rational generation. Integral localization proves saturation at the tested primes; an exact cyclotomic determinantal-ideal certificate then closes the complete $\rho/\mathcal{B}_1$ equation obstruction away from 2, 3, and 17. All numerical claims are accompanied by exact finite certificates; the certificate contract is described in Appendix A.

Organization. Section 2 fixes notation and the rational characters. Section 3 decomposes the relation modules. Section 4 proves the local circuit statement and gives the exhaustive census. Section 5 proves rational generation, and Section 6 determines the cubic toric layer. Section 7 treats the primes 2, 3, 17. Section 8 proves the complete primary classifications at 137, 219 097, and 3 288 036 131. Section 9 gives the global localized theorem for the single complete $\rho/\mathcal{B}_1$ component. Open integral and higher-degree questions are separated in Section 10.

2 The locked triple and its common symmetry

All subspace dimensions in this paper are vector dimensions. Thus a block $B \in \mathcal{B}_i$ is a four-dimensional subspace of $\mathbb{F}_2^8$ and contains 15 projective points. The free block and point lattices are

$$M_i^{\mathbb{Z}} = \mathbb{Z}[\mathcal{B}_i], \quad P^{\mathbb{Z}} = \mathbb{Z}[\mathcal{P}].$$

We use superscripts $\mathbb{Q}$ and k for scalar extension. The integral and rational relation modules are

$$K_i^{\mathbb{Z}} = \ker(A_i : M_i^{\mathbb{Z}} \rightarrow P^{\mathbb{Z}}), \quad K_i = K_i^{\mathbb{Z}} \otimes \mathbb{Q}.$$

Since A_i has full row rank,

$$\dim_{\mathbb{Q}} K_i = 66\,929 - 255 = 66\,674.$$

2.1 Immutable identification of the three designs

The finite objects are reconstructed by the companion repository *FCIG-Common-Marked-Lattice-Designs*, locked at Git commit 40b3e716834e5c02e907cdadc067b3081d7f3227. In canonical block order, the SHA-256 identifiers are

	canonical-key stream	point-mask stream
$\mathcal{B}_1$	3F267CCC...30B676B	8220960A...655BBDF
$\mathcal{B}_2$	773AAEE7...D8EFB5	01D77747...651261
$\mathcal{B}_3$	D2829B5A...3F723	48386150...56E46A

The unabbreviated hashes and the common 66 929-block count are recorded in the source lock; the union identifier is

$$1F09C96E...DCCA4E.$$

These identifiers make the subject of every finite computation independent of a filename or mutable checkout.

The common group is generated by elements a, b, c satisfying

$$a^{17} = b^4 = c^3 = 1, \quad bab^{-1} = a^4, \quad [a, c] = [b, c] = 1.$$

Set $F = \langle a, b \rangle \cong C_{17} \rtimes C_4$. The action of $\Gamma = F \times \langle c \rangle$ preserves every block collection and the point set, so every incidence map A_i is Γ -equivariant.

2.2 The eight rational types

The rational irreducibles can be constructed without choosing a matrix model. The quotient $F/C_{17} \cong C_4$ has the three rational types $\mathbf{1}, \varepsilon, \iota$, of dimensions 1, 1, 2. Here ε is the order-two character and ι is the rational form of the pair of order-four characters. The group C_3 has rational types $\mathbf{1}, \omega$, where ω is the two-dimensional rational form of the two nontrivial complex characters.

For a nontrivial complex character χ of C_{17} , the C_4 action has an orbit of size four, and $\text{Ind}_{C_{17}}^F \chi$ has complex degree four. The four such induced characters form one Galois orbit. Their rational form is denoted ρ and has dimension 16. Tensoring with the nontrivial C_3 -orbit gives $\rho\omega$, of dimension 32.

Thus the rational types and dimensions are

V	$\mathbf{1}$	ε	ι	ω	$\varepsilon\omega$	$\iota\omega$	ρ	$\rho\omega$
$\dim_{\mathbb{Q}} V$	1	1	2	2	2	4	16	32.

For completeness, over $\mathbb{C}$ the first six rational types account for all 12 linear characters of Γ . The type ρ accounts for four degree-four characters and $\rho\omega$ for the remaining eight. Consequently

$$12 \cdot 1^2 + 12 \cdot 4^2 = 204,$$

so the list exhausts the complex irreducibles and hence the rational types.

Proposition 2.1 (Point module). *The rational point permutation module is*

$$\mathbb{Q}[\mathcal{P}] \cong 3\mathbf{1} \oplus 2\varepsilon \oplus 3\omega \oplus 2\varepsilon\omega \oplus 5\rho \oplus 5\rho\omega.$$

In particular, neither ι nor $\iota\omega$ occurs.

Proof. For each of the 24 conjugacy classes, the permutation character equals the number of fixed projective points. Exact inner products with the Galois sums constructed above give the displayed multiplicities. Their dimension-weighted sum is

$$3 + 2 + 6 + 4 + 80 + 160 = 255,$$

which checks the full point dimension. The class values and inner products are included in the electronic representation certificate described in Appendix A. $\square$

Remark 2.2. *Only the locked finite triple is considered here. No family classification is inferred from its common group or its character table.*

3 Rational relation modules

Because $A_i : M_i^{\mathbb{Q}} \rightarrow P^{\mathbb{Q}}$ is a surjective $\mathbb{Q}\Gamma$ -map, semisimplicity gives

$$[K_i] = [M_i^{\mathbb{Q}}] - [P^{\mathbb{Q}}]$$

in the rational representation ring. Exact character inner products therefore determine K_i from the block and point permutation characters.

Proposition 3.1 (Complete relation-module table). *The multiplicities of the eight rational types in K_1, K_2, K_3 are as follows.*

V	K_1	K_2	K_3	$\dim_{\mathbb{Q}} V$
$\mathbf{1}$	343	354	355	1
ε	341	352	351	1
ι	314	304	302	2
ω	339	349	353	2
$\varepsilon\omega$	338	347	349	2
$\iota\omega$	314	304	302	4
ρ	1312	1314	1310	16
$\rho\omega$	1305	1304	1306	32
<i>weighted total</i>	66 674	66 674	66 674	

In particular, K_1, K_2, K_3 are pairwise nonisomorphic as $\mathbb{Q}\Gamma$ -modules.

Proof. Subtract the point multiplicities in proposition 2.1 from the three exact block permutation characters. The resulting table is the one displayed. For example, the two large-type contributions are

$$16(1312, 1314, 1310) \quad \text{for } \rho, \quad 32(1305, 1304, 1306) \quad \text{for } \rho\omega.$$

Summing every row with its rational dimension gives 66 674 in each column, agreeing with the incidence rank.

The three entries in every row are not all equal; more strongly, each pair of columns differs in at least one multiplicity, and inspection shows that the three configurations are separated in every rational type. Uniqueness of semisimple multiplicities proves pairwise nonisomorphism. $\square$

Remark 3.2 (What the table separates). *The trivial multiplicities 343, 354, 355 are the dimensions of the invariant relation spaces and can also be recovered from block-orbit counts. The structural conclusion is the simultaneous decomposition of the full 66 674-dimensional kernels. It should not be confused with equality of incidence images: all three maps have the same full rational image $P^{\mathbb{Q}}$, while their kernels are inequivalent.*

4 Minimum subspace trades

A point trade in $\mathcal{B}_i$ is an integral relation written with disjoint positive and negative supports,

$$T = T^+ - T^- = \sum_{B \in T^+} B - \sum_{B \in T^-} B \in K_i^{\mathbb{Z}}.$$

Its volume is $|T^+| = |T^-|$, and its total support is $|T^+| + |T^-|$. This distinction matters here: the minimum $T_2(1, 4, 8)$ trades have volume 3 and total support 6.

4.1 The opposite-regulus model

Choose vector subspaces $Z < U < \mathbb{F}_2^8$ with $\dim Z = 2$ and $\dim U = 6$. A four-space B satisfying $Z < B < U$ projects to a projective line B/Z in $\text{PG}(U/Z) \cong \text{PG}(3, 2)$. Three pairwise skew lines form one regulus of the hyperbolic quadric $Q^+(3, 2)$; the three lines meeting all of them form its opposite regulus. Lifting the two reguli gives two triples of four-spaces

$$\{B_1, B_2, B_3\}, \quad \{B'_1, B'_2, B'_3\},$$

with the same point-incidence sum.

Lemma 4.1 (Primitive regulus circuit). *Let N be the 255×6 point-incidence matrix of the six lifted blocks, ordered by the two reguli. Then*

$$\text{rank}_{\mathbb{Q}} N = 5, \quad \ker_{\mathbb{Z}} N = \mathbb{Z}(1, 1, 1, -1, -1, -1).$$

Every five columns are independent, and the nonzero Smith factors of N are 1, 1, 1, 1, 1. Hence the displayed trade is a primitive unimodular matroid circuit and a Graver element.

Proof. In the quotient $\text{PG}(3, 2)$, each point of the hyperbolic quadric lies on one line from each regulus, giving the displayed kernel vector. Direct row reduction of the 15-point quotient incidence matrix has rank five, and deleting any one line leaves five independent columns. Lifting through $Z < U$ repeats rows but introduces no new column relation. The active Smith form is $\text{diag}(1, 1, 1, 1, 1)$, so the one-dimensional integral kernel is primitive and generated by the sign vector above. A primitive support-minimal relation is a Graver element. $\square$

Krotov's minimum-trade theorem applies with

$$0 \leq t = 1 < k = 4 < v - t = 7.$$

Its support bound specializes to $\prod_{j=0}^1 (1 + 2^j) = 6$, hence volume three, and its $t = 1$ classification identifies every trade attaining this bound with the opposite-regulus construction above [8]. We use that classification for exhaustiveness, while lemma 4.1 supplies the integral circuit statement.

4.2 Exact census

Let $\mathcal{C}_i$ be the set of unoriented minimum trades wholly contained in $\mathcal{B}_i$. Exhaustive enumeration first chooses (Z, U) , constructs the two reguli in U/Z , and then tests membership of all six lifts in $\mathcal{B}_i$. Canonical sorting removes repeated presentations, and the common group then partitions the result into orbits.

Proposition 4.2 (Minimum-trade census). *The numbers of minimum trades and their Γ -orbit counts are*

	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$ \mathcal{C}_i $	2 271 676	2 213 026	2 216 868
$ \mathcal{C}_i/\Gamma $	11 274	11 001	11 033.

Every trade in the census has coefficient gcd one and active Smith form 1^5 . Thus every local arithmetic multiplicity is one.

Proof. Krotov's $t = 1$ classification reduces the search to the (Z, U) -regulus templates, so the enumeration is exhaustive rather than a bounded-support sample. Three independent passes—template generation, direct six-block membership, and orbit closure—give the displayed totals. For each accepted template, lemma 4.1 gives the coefficient gcd and Smith form. The frozen counts, orbit representatives, and replay hashes form the census certificate of Appendix A. $\square$

Remark 4.3. *The configurations are separated by how many minimum circuits they contain and how those circuits are organized into group orbits. They are not separated by a different six-block local arithmetic type. Nothing in this section classifies minimum $T_2(2, 4, 8)$ trades; for that parameter the same support formula gives 30, or volume 15.*

5 Full rational generation

Let $C_i \subseteq K_i$ be the rational span of all trades in $\mathcal{C}_i$. We prove $C_i = K_i$ by testing the quotient in every irreducible component. This section states both the abstract argument and the exact finite certificate to which it is applied.

5.1 Component equation matrices

Let k be a splitting field for Γ whose characteristic does not divide 204, and let V be an absolutely irreducible $k\Gamma$ -module. Choose one representative in each block orbit and write its stabilizer as H_j . Frobenius reciprocity gives

$$\mathrm{Hom}_{\Gamma}(k[\mathcal{B}_i], V) \cong \bigoplus_j V^{H_j}.$$

The analogous point-orbit decomposition gives $\mathrm{Hom}_{\Gamma}(k[\mathcal{P}], V)$. Put

$$m_{B_i}(V) = \dim_k \mathrm{Hom}_{\Gamma}(k[\mathcal{B}_i], V), \quad m_P(V) = \dim_k \mathrm{Hom}_{\Gamma}(k[\mathcal{P}], V).$$

Choose minimum-trade orbit seeds $c_1, \dots, c_s$. Evaluation on these seeds defines the component equation map

$$E_{i,V} : \text{Hom}_\Gamma(k[\mathcal{B}_i], V) \longrightarrow V^s, \quad \phi \longmapsto (\phi(c_1), \dots, \phi(c_s)).$$

Once bases of all fixed spaces have been frozen, this is an ordinary matrix over k . Killing one seed kills its entire Γ -orbit, because $\phi(gc) = g\phi(c)$.

Every map $k[\mathcal{P}] \rightarrow V$, pulled back along incidence, kills all circuits. Since the incidence map is surjective, this pullback is injective. Therefore

$$\text{rank } E_{i,V} \leq m_{B_i}(V) - m_P(V).$$

Lemma 5.1 (Component rank criterion). *If*

$$\text{rank } E_{i,V} = m_{B_i}(V) - m_P(V)$$

for every absolutely irreducible V , then $C_i = K_i$.

Proof. The kernel of $E_{i,V}$ is $\text{Hom}_\Gamma(k[\mathcal{B}_i]/C_i, V)$. Equality in the rank bound says that this kernel has dimension $m_P(V)$, exactly the dimension of the incidence pullback. Hence

$$\text{Hom}_\Gamma(k[\mathcal{B}_i]/C_i, V) = \text{Hom}_\Gamma(k[\mathcal{P}], V)$$

for every V . The induced incidence map $k[\mathcal{B}_i]/C_i \rightarrow k[\mathcal{P}]$ is surjective. Its source and target have the same semisimple multiplicities, so it is an isomorphism. Its kernel is $(K_i/C_i) \otimes k$, which is therefore zero. Characteristic-zero scalar extension gives $K_i = C_i$. $\square$

5.2 The exact rank certificate

The primes 409 and 613 are congruent to 1 modulo 204. Thus both fields split Γ . For each of the three designs and each of the eight Galois-orbit representatives, the frozen component matrix has a square minor of order

$$m_{B_i}(V) - m_P(V)$$

whose determinant is nonzero modulo 409. These 24 minors prove the required characteristic-zero ranks. Reconstructing the same 24 ranks modulo 613 is an independent robustness check rather than an additional premise.

The seed stream is round-robin across all 69 two-space orbits. This condition is structural: a lexicographic stream can omit invariant mass, whereas the frozen stream meets every orbit stratum. The certificate stores the orbit/stabilizer input, fixed-space bases, selected row and column indices, determinant residue, and a negative control obtained by removing one essential row or column.

Theorem 5.2 (Minimum trades span the rational kernel). *For $i = 1, 2, 3$,*

$$C_i = K_i.$$

Moreover, the Γ -spans of deterministic prefixes of respectively

$$1\ 143, \quad 1\ 170, \quad 648$$

minimum-trade seeds already equal K_1, K_2, K_3 .

Proof. The nonzero minors modulo 409 attain the upper bound in lemma 5.1 for one absolute representative in every rational Galois orbit. All circuit vectors are rational, so Galois conjugation transports each rank statement to the other absolute constituents in its orbit. The criterion gives $C_i = K_i$.

For each design, take the largest prefix length used by any of its eight component certificates. Every component has already attained its target rank at that common prefix, giving the three stated bounds. $\square$

Remark 5.3. *The theorem is not a consequence of the census alone. It depends on attaining the exact rank in every irreducible component. Conversely, the minor at 409 is sufficient for the characteristic-zero transfer; the computation at 613 checks the frozen construction independently.*

6 The first toric layer

Fix i and a field k . Let

$$S_i = k[x_B : B \in \mathcal{B}_i]$$

and define the incidence monomial map

$$\pi_i : S_i \longrightarrow k[y_p : p \in \mathcal{P}], \quad x_B \longmapsto \prod_{p \in \text{PG}(B)} y_p.$$

The incidence toric ideal is $I_i = \ker \pi_i$. Since every block contains 15 points, I_i is homogeneous for total degree.

Lemma 6.1 (No linear or quadratic moves). *The ideal I_i has no nonzero binomial of degree one or two.*

Proof. Distinct blocks have distinct point-incidence columns, excluding degree one. Suppose a quadratic relation remains after cancelling common variables, and choose a block B from one side. If that side contains B^2 , its incidence vector has scalar product at least 30 with the column of B . Otherwise it contains B once and the scalar product is at least 15. A different four-space meets B in at most 7 projective points, so the two blocks on the opposite side contribute at most 14. Both cases are impossible. $\square$

Lemma 6.2 (Squarefree cubic classification). *Every nonzero cubic binomial in I_i , after cancellation, is the binomial of a minimum opposite-regulus trade.*

Proof. If a block B is repeated on one side, its scalar product with that side is at least 30. The three distinct blocks on the other side contribute at most $3 \cdot 7 = 21$. Thus both cubic monomials are squarefree. A block shared by the two sides would cancel to a nonzero quadratic, contradicting lemma 6.1. Hence the binomial has three blocks in each leg and total support six. It is a minimum $T_2(1, 4, 8)$ trade, and Krotov's $t = 1$ classification identifies it with an opposite-regulus trade [8]. $\square$

For a monomial M , let

$$\mathcal{F}(M) = \{N : \pi_i(N) = \pi_i(M)\}$$

be its multidegree fiber.

Lemma 6.3 (Two-element fibers). *If $M = x_{B_1}x_{B_2}x_{B_3}$ is one leg of a minimum trade and N is the other leg, then*

$$\mathcal{F}(M) = \{M, N\}.$$

Proof. Every monomial in the fiber has degree three. By lemmas 6.1 and 6.2, a second monomial is squarefree, shares no block with M , and forms a minimum regulus trade with it.

The three blocks in M recover

$$Z = B_1 \cap B_2 \cap B_3, \quad U = B_1 + B_2 + B_3.$$

Their images in $\text{PG}(U/Z) \cong \text{PG}(3, 2)$ are three skew lines. Those lines have a unique opposite regulus, which lifts to the three blocks of N . Thus no third monomial occurs. $\square$

Theorem 6.4 (Complete cubic layer). *For each i , the primitive cubics of I_i are exactly the minimum opposite-regulus binomials. Every one is indispensable. Their numbers are*

$$2\,271\,676, \quad 2\,213\,026, \quad 2\,216\,868$$

for $i = 1, 2, 3$, respectively. Equivalently,

$$\beta_{0,3}(I_i) = \beta_{1,3}(S_i/I_i) = |\mathcal{C}_i|.$$

Proof. Lemma 6.2 identifies all cubics, and lemma 6.3 shows that each has a two-element fiber. The indispensable-fiber criterion of Aoki–Takemura–Yoshida implies that the corresponding binomial occurs, up to scalar, in every Markov basis [1]. Since lemma 6.1 excludes lower degrees, these cubics are minimal generators. The exact counts are those of proposition 4.2. $\square$

Remark 6.5 (Higher degrees). *The theorem determines the first nonzero graded layer. It does not prove that I_i is generated by cubics, nor does it determine a complete Markov or Graver basis.*

7 Integral relation lattices

Let

$$L_i = \langle T : T \in \mathcal{C}_i \rangle_{\mathbb{Z}} \subseteq K_i^{\mathbb{Z}}, \quad \Theta_i = K_i^{\mathbb{Z}}/L_i.$$

The rational theorem determines the rank but not the index of L_i . This distinction is the source of the integral phenomenon below.

Lemma 7.1 (Finiteness and local saturation). *The group Θ_i is finite. Let p be a prime. If the image of $L_i \otimes_{\mathbb{Z}_p}$ spans $(K_i^{\mathbb{Z}} \otimes_{\mathbb{Z}_p})/p(K_i^{\mathbb{Z}} \otimes_{\mathbb{Z}_p})$, then $\Theta_i[p^{\infty}] = 0$.*

Proof. The lattice $K_i^{\mathbb{Z}}$ is primitive in $\mathbb{Z}[\mathcal{B}_i]$: its quotient is the image of A_i , a subgroup of the free group $\mathbb{Z}[\mathcal{P}]$. By theorem 5.2, L_i and $K_i^{\mathbb{Z}}$ have the same rational span and hence the same rank. Their quotient is finite.

Put $M = K_i^{\mathbb{Z}} \otimes_{\mathbb{Z}_p}$ and $N = L_i \otimes_{\mathbb{Z}_p}$. The spanning hypothesis is $N + pM = M$. Nakayama's lemma gives $N = M$, so the localized quotient vanishes. $\square$

7.1 The prime 2

Proposition 7.2 (Binary saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[2^{\infty}] = 0.$$

Proof. In each design, an explicit set of 66 674 minimum trades is linearly independent in $\mathbb{F}_2[\mathcal{B}_i]$. Primitivity of $K_i^{\mathbb{Z}}$ makes $K_i^{\mathbb{Z}}/2K_i^{\mathbb{Z}}$ a 66 674-dimensional subspace of that ambient permutation module. The selected trades lie in this subspace and already have its full dimension. Apply lemma 7.1. $\square$

This statement concerns liftable integral relations, not the complete modular null-design space. The distinction is especially visible at the next prime and is consistent with the modular minimum-support setting of [9].

7.2 The prime 3: the liftable kernel

Let $\alpha : \mathbb{Z}[\mathcal{B}_i] \rightarrow \mathbb{Z}$ be augmentation, $\alpha(\sum n_B B) = \sum n_B$. Summing all point coordinates of an integral incidence relation gives

$$15\alpha(x) = 0.$$

Thus α vanishes on $K_i^{\mathbb{Z}}$, even though its reduction modulo 3 is not a consequence of the incidence equations.

Lemma 7.3 (Augmentation–Bockstein identification). *For every i ,*

$$K_i^{\mathbb{Z}}/3K_i^{\mathbb{Z}} = \ker_{\mathbb{F}_3} \begin{bmatrix} A_i \\ \alpha \end{bmatrix}.$$

Proof. The locked Smith data have exactly one factor divisible by 3, namely 120, in the 255-row incidence presentation. In particular,

$$\text{rank}_{\mathbb{F}_3} A_i = 254, \quad \text{rank}_{\mathbb{F}_3} \begin{bmatrix} A_i \\ \alpha \end{bmatrix} = 255.$$

The modular kernel of A_i therefore has dimension 66 675, one larger than the reduction of the primitive rank-66 674 integral kernel. The augmented kernel has dimension 66 674 and contains that reduction by the preceding augmentation identity. Equality follows. Equivalently, the missing modular direction is the one-dimensional Tor_1 contribution produced by the factor 120. $\square$

Write $\Gamma = F \times C_3$, where $|F| = 68$, and put

$$R_3 = \mathbb{F}_3[C_3] \cong \mathbb{F}_3[u]/(u^3), \quad u = c - 1.$$

The ideal (u) is the radical of $\mathbb{F}_3\Gamma$, while $\mathbb{F}_3F$ is semisimple. Let $M_i^{(3)}$ denote the liftable kernel in lemma 7.3, and let $\overline{M}_i^{(3)} = M_i^{(3)}/uM_i^{(3)}$ be its semisimple top.

The exact component certificates decompose this top into four Frobenius families, whose Wedderburn dimensions account for all 68 dimensions of $\mathbb{F}_3F$. The images of minimum trades attain total top dimensions

	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$\dim_{\mathbb{F}_3} \overline{M}_i^{(3)}$	22 304	22 338	22 270.

Each component rank is accompanied by a maximal minor and a rank-loss control.

Proposition 7.4 (Ternary saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[3^\infty] = 0.$$

Proof. By lemma 7.3, the module tested by the certificates is exactly the reduction of the integral kernel, not the one-dimension-larger modular kernel. The component ranks say that the minimum trades generate every simple summand of $\overline{M}_i^{(3)}$. Hence

$$M_i^{(3)} = L_i^{(3)} + uM_i^{(3)}.$$

The ideal (u) is nilpotent, so Nakayama’s lemma gives $M_i^{(3)} = L_i^{(3)}$. Applying the local criterion to the integral lattices proves the assertion. $\square$

7.3 The prime 17: a free cyclic kernel

The normal subgroup C_{17} acts freely on points and blocks. There are 15 point orbits and 3937 block orbits. Put

$$R_{17} = \mathbb{F}_{17}[C_{17}] \cong \mathbb{F}_{17}[u]/(u^{17}), \quad u = a - 1.$$

The two permutation modules are free R_{17} -modules of ranks 3937 and 15. The coinvariant incidence map is surjective. Nakayama therefore makes the full R_{17} -map surjective, and its kernel $M_i^{(17)}$ is free of rank

$$3937 - 15 = 3922.$$

The quotient $H = \Gamma/C_{17} \cong C_4 \times C_3$ is semisimple in characteristic 17. Its eight simple components have total Wedderburn mass $12 = |H|$. In every component, exact fixed-space matrices show that the minimum trades span

$$M_i^{(17)}/uM_i^{(17)}.$$

Proposition 7.5 (Seventeen-adic saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[17^\infty] = 0.$$

Proof. The eight component ranks generate the complete semisimple top of the free R_{17} -kernel. Thus $M_i^{(17)} = L_i^{(17)} + uM_i^{(17)}$. Nakayama over the local ring R_{17} gives equality, and lemma 7.1 lifts the conclusion to the 17-adic lattices. $\square$

7.4 What the small components exclude

Away from $2 \cdot 3 \cdot 17$, the six rational types of dimensions at most four admit integral component models. For each of the resulting 18 maps, two exact maximal minors have the gcd shown below.

V	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$\mathbf{1}$	24	6	24
ε	8	2	128
ι	1024	4096	128
ω	4	216	8
$\varepsilon\omega$	108	12	2
$\iota\omega$	33 554 432	16 777 216	49 152

Every displayed number is supported on $\{2, 3, 17\}$. Bezout combination of the corresponding minors makes each maximal-minor ideal a unit after these primes are inverted.

Proposition 7.6 (Small-component exclusion). *No prime outside $\{2, 3, 17\}$ occurs in any of the 18 component cokernels belonging to*

$$\mathbf{1}, \varepsilon, \iota, \omega, \varepsilon\omega, \iota\omega.$$

This is not a support theorem for Θ_i . The six large maps $\rho/\mathcal{B}_i$ and $\rho\omega/\mathcal{B}_i$ require separate complete-equation-map analysis. Section 8 resolves the selected primes 137, 219097, and 3288036131; Section 9 then closes the single $\rho/\mathcal{B}_1$ component globally away from 2, 3, and 17.

8 Complete saturation at three selected good primes

We first record the exact passage from complete component equations to the integral quotient.

Lemma 8.1 (Good-prime component detection). *Let $p \nmid |\Gamma|$, suppose $A_i \otimes \mathbb{F}_p$ is surjective, and let $k/\mathbb{F}_p$ split Γ . Put*

$$Q_i = (K_i^{\mathbb{Z}}/L_i) \otimes k.$$

For every absolutely irreducible $k\Gamma$ -module V , the structural rank deficiency of the complete component equation map satisfies

$$(m_{B_i}(V) - m_P(V)) - \text{rank } E_{i,V} = \dim_k \text{Hom}_{k\Gamma}(Q_i, V).$$

Consequently, if every complete component map has its structural target rank, then $\Theta_i \otimes \mathbb{F}_p = 0$, and hence $\Theta_i[p^\infty] = 0$.

Proof. Reduction gives an exact sequence

$$0 \longrightarrow Q_i \longrightarrow k[\mathcal{B}_i]/(L_i \otimes k) \longrightarrow k[\mathcal{P}] \longrightarrow 0.$$

Maschke's theorem makes $k\Gamma$ semisimple. Applying $\text{Hom}_{k\Gamma}(-, V)$ gives

$$0 \longrightarrow \text{Hom}_{k\Gamma}(k[\mathcal{P}], V) \longrightarrow \ker E_{i,V} \longrightarrow \text{Hom}_{k\Gamma}(Q_i, V) \longrightarrow 0.$$

Taking dimensions gives the displayed deficiency formula. If every absolute type vanishes after a splitting extension, faithful scalar extension gives $Q_i = 0$. Finiteness of Θ_i and Nakayama's lemma then give $\Theta_i[p^\infty] = 0$. $\square$

8.1 The prime 137

Here $137 \nmid 204$, every projector denominator is a unit, and the incidence Smith form has no 137-factor. For $\rho/\mathcal{B}_1$, the block multiplicity space has dimension 1317, while the point-induced subspace has dimension 5, so the structural equation rank is

$$1317 - 5 = 1312.$$

The complete circuit-equation matrix has a certified 1312×1312 minor with

$$\det = 56 \pmod{137} \quad \text{at } \zeta_{17} = 122.$$

Theorem 8.2 (Complete $\rho/\mathcal{B}_1$ contribution above 137). *The complete $\rho/\mathcal{B}_1$ equation matrix has structural rank 1312 at every prime above 137.*

Proof. The displayed minor is nonzero at $\zeta_{17} = 122$; the exact certificates at the other fifteen roots likewise contain nonzero order-1312 minors. $\square$

For $\rho\omega$, put

$$k_2 = \mathbb{F}_{137}[y]/(y^2 + y + 1).$$

If r runs over the sixteen order-17 roots in $\mathbb{F}_{137}$, then

$$\alpha_r = r^6 y^2, \quad \alpha_r^3 = r, \quad \alpha_r^{17} = y$$

has order 51, and the Frobenius pairs $\{\alpha_r, \alpha_r^{137}\}$ represent the sixteen degree-two primes of $\mathbb{Z}[\zeta_{51}]$ above 137. The six large-map certificates are

map	target rank	columns	residue degrees	nonzero minors
$\rho/\mathcal{B}_1$	1312	1317	1^{16}	16/16
$\rho\omega/\mathcal{B}_1$	1305	10480	2^{16}	16/16
$\rho/\mathcal{B}_2$	1314	5276	1^{16}	16/16
$\rho\omega/\mathcal{B}_2$	1304	10472	2^{16}	16/16
$\rho/\mathcal{B}_3$	1310	5260	1^{16}	16/16
$\rho\omega/\mathcal{B}_3$	1306	10488	2^{16}	16/16.

For quadratic residues, multiplication by $a + by$ is expanded as

$$\begin{bmatrix} a & -b \\ b & a - b \end{bmatrix}.$$

An independent parser uses Frobenius-conjugate roots, transposes every selected square matrix, reproduces all 80 determinant residues for the last five maps, and obtains zero after duplicating a field column.

Theorem 8.3 (Complete 137-primary quotients). *For the three locked configurations,*

$$\Theta_1[137^\infty] = \Theta_2[137^\infty] = \Theta_3[137^\infty] = 0.$$

Proof. Proposition 7.6 excludes 137 from all eighteen small component cokernels. The table supplies structural rank for all six large maps at every localization. Apply lemma 8.1. $\square$

8.2 The prime 219 097

Since

$$219\,097 - 1 = 2^3 \cdot 3^2 \cdot 17 \cdot 179, \quad 219\,097 \equiv 1 \pmod{51},$$

both cyclotomic orders split completely. The complete $\rho/\mathcal{B}_1$ equation matrix has a certified full-rank minor with

$$\det = 213269 \pmod{219097} \quad \text{at } \zeta_{17} = 91085.$$

The complete large-map screen is

map	target rank	localizations	nonzero minors
$\rho/\mathcal{B}_1$	1312	16	16
$\rho\omega/\mathcal{B}_1$	1305	32	32
$\rho/\mathcal{B}_2$	1314	16	16
$\rho\omega/\mathcal{B}_2$	1304	32	32
$\rho/\mathcal{B}_3$	1310	16	16
$\rho\omega/\mathcal{B}_3$	1306	32	32.

For the last five maps, an independent parser enumerates the roots from a primitive generator, reconstructs the transpose of every selected square matrix, reproduces all 128 determinants, and obtains zero in all 128 duplicate-column controls.

Theorem 8.4 (Complete 219 097-primary quotients). *For the three locked configurations,*

$$\Theta_1[219097^\infty] = \Theta_2[219097^\infty] = \Theta_3[219097^\infty] = 0.$$

Proof. Proposition 7.6 excludes 219 097 from all eighteen small components, and the complete large-map screen supplies structural rank for the remaining six. Apply lemma 8.1. $\square$

8.3 The prime 3 288 036 131

Put $p = 3\,288\,036\,131$. Then

$$p - 1 = 2 \cdot 5 \cdot 17 \cdot 2243 \cdot 8623, \quad p \equiv 1 \pmod{17}, \quad p \equiv 2 \pmod{3}.$$

Thus Φ_{17} has sixteen linear factors and Φ_{51} has sixteen irreducible quadratic factors. The complete $\rho/\mathcal{B}_1$ equation matrix has a certified full-rank minor with

$$\det = 1\,471\,654\,912 \pmod{p} \quad \text{at } \zeta_{17} = 1\,120\,906\,769.$$

For the $\rho\omega$ -maps, use

$$k_2 = \mathbb{F}_p[y]/(y^2 + y + 1), \quad \alpha_r = r^6 y^2,$$

where r runs through the order-17 roots. The complete screen is

map	target rank	residue degree	localizations	nonzero minors
$\rho/\mathcal{B}_1$	1312	1	16	16
$\rho\omega/\mathcal{B}_1$	1305	2	16	16
$\rho/\mathcal{B}_2$	1314	1	16	16
$\rho\omega/\mathcal{B}_2$	1304	2	16	16
$\rho/\mathcal{B}_3$	1310	1	16	16
$\rho\omega/\mathcal{B}_3$	1306	2	16	16.

The independent route finds the least primitive generator 2, uses Frobenius-conjugate quadratic representatives, transposes every selected square matrix, reproduces all 80 determinant residues for the last five maps, and obtains zero in all 80 duplicate-column controls.

Theorem 8.5 (Complete 3 288 036 131-primary quotients). *For the three locked configurations,*

$$\Theta_1[p^\infty] = \Theta_2[p^\infty] = \Theta_3[p^\infty] = 0.$$

Proof. Proposition 7.6 excludes p from all eighteen small components, and the complete large-map screen supplies structural rank for the remaining six. Apply lemma 8.1. $\square$

Remark 8.6 (Exact boundary). *The three theorems classify the complete primary quotients only at the displayed rational primes. A finite list of local screens does not determine the support or Smith factors of the global groups Θ_i . The next section gives a global determinantal-ideal certificate for the single $\rho/\mathcal{B}_1$ equation module.*

9 A global localized theorem for the $\rho/\mathcal{B}_1$ component

Put

$$R = \mathbb{Z}[\zeta], \quad \zeta = \zeta_{17}, \quad R_S = R \left[\frac{1}{2 \cdot 3 \cdot 17} \right].$$

The power basis $1, \zeta, \dots, \zeta^{15}$ is an integral basis of R [14]. Let

$$E = E_{\rho,1}^{\text{full}}$$

be the matrix whose rows are all complete minimum-circuit equations and whose 1317 columns are the fixed $\rho/\mathcal{B}_1$ multiplicity coordinates. The five-dimensional point-induced subspace lies in the kernel of every circuit equation. Hence

$$\text{rank } E \leq 1317 - 5 = 1312$$

after every scalar extension. We write $I_{1312}(E) \subseteq R$ for the ideal generated by all order-1312 minors of this complete equation matrix.

9.1 Exact lift and determinant reconstruction

Lemma 9.1 (Common cyclotomic lift). *The 1317 source columns and every complete minimum-circuit equation for E have one exact realization over R in the power basis above. Under a residue homomorphism sending ζ to an element of order 17, this realization reduces entrywise to the corresponding complete finite component equations.*

Proof. The fixed-space columns are normalized as exponent vectors modulo 17. Every representation entry is a signed sum of those powers, reduced by

$$1 + \zeta + \cdots + \zeta^{16} = 0.$$

Circuit equations are formed from the same six signed block columns before specialization. Thus evaluation at a seventeenth root commutes with column construction and row formation. $\square$

Lemma 9.2 (Certified signed CRT reconstruction). *Let B be either square matrix in proposition 9.3, and write*

$$\det B = \sum_{j=0}^{15} c_j \zeta^j.$$

Hadamard’s inequality and Fourier inversion give the certified bound

$$\max_j |c_j| \leq \frac{32}{17} \max_{\sigma: R \hookrightarrow \mathbb{C}} |\sigma(\det B)|.$$

If the product M of the recorded split CRT primes is larger than twice this bound, the signed coefficients c_j are uniquely determined by their residues modulo M .

Proof. For each embedding, Hadamard bounds the determinant by the square root of the product of the certified row energies. Applying the inverse seventeenth-root transform and eliminating the coefficient of ζ^{16} gives the displayed bound. Uniqueness follows coordinatewise in the signed interval $(-M/2, M/2)$. $\square$

Proposition 9.3 (Two exact order-1312 minors). *The complete equation matrix E contains two explicitly specified 1312×1312 submatrices, using different equation-row bases and different 1312-column selections, with determinants $\delta_G, \delta_R \in R$. Their frozen exact reconstruction data are*

<i>minor</i>	<i>entries</i>	<i>direct check</i>	<i>CRT primes</i>	<i>bound bits</i>	<i>modulus bits</i>	<i>norm bits</i>
δ_G	7830	89 mod 103	96	2937	2967	4833
δ_R	7819	107 mod 137	93	2872	2874	2003

The largest reconstructed coefficient has respectively 304 and 128 bits.

Proof. For δ_G , a complete equation-row basis at $(103, \zeta - 8)$, together with its specified column selection, gives determinant 89. Its exact sparse lift has 7830 nonzero entries. For δ_R , reverse canonical traversal of the complete circuit equations supplies a different row basis; with columns $0, \dots, 1311$, its determinant is 107 at $(137, \zeta - 122)$, and its exact lift has 7819 nonzero entries.

Standalone binary parsers directly reconstruct and check these two diagnostic determinants. The public reconstruction verifier separately audits the frozen CRT prime lists, every recorded coefficient congruence, the row-energy bounds, signed uniqueness, the exact cyclotomic norms, and mutation controls. The CRT determinant-generation transcript itself is a source-admitted certificate rather than a clean-room recomputation of all 189 modular determinants. $\square$

9.2 The determinantal-ideal certificate

Lemma 9.4 (Base change for determinantal ideals). *For a matrix M over a commutative ring T , a ring map $T \rightarrow T'$, and an integer r ,*

$$I_r(M)T' = I_r(M \otimes_T T').$$

If $I_r(M)T' = T'$, then M has rank at least r over every local ring of T' .

Proof. Both statements follow by applying the ring map to the determinants of all $r \times r$ submatrices. If the extended ideal is the unit ideal, at least one such determinant is a unit after localization at each prime. $\square$

Lemma 9.5 (Multiplication-lattice certificate). *For $\alpha \in R$, let $M(\alpha) \in \text{Mat}_{16}(\mathbb{Z})$ be multiplication by α in the integral power basis. Then the columns of*

$$[M(\alpha) \ M(\beta)] \in \text{Mat}_{16 \times 32}(\mathbb{Z})$$

span $\alpha R + \beta R$, and its nonzero Smith entries are the additive invariant factors of $R/(\alpha R + \beta R)$.

Proof. The columns of $M(\alpha)$ are $\alpha, \alpha\zeta, \dots, \alpha\zeta^{15}$ in coordinates. Concatenation adds the two principal-ideal lattices, and Smith normal form computes their quotient in the ambient free abelian group. $\square$

Theorem 9.6 (Global localized structural rank of the $\rho/\mathcal{B}_1$ equations). *For the complete minimum-circuit equation matrix E ,*

$$I_{1312}(E)R_S = R_S.$$

Consequently $E \otimes_R R_S$ has rank exactly 1312 at every prime of R_S , and the residual $\rho/\mathcal{B}_1$ component equation obstruction vanishes there.

Proof. The independently audited multiplication-lattice Smith form for $[M(\delta_G) \ M(\delta_R)]$ has diagonal

$$\underbrace{17, \dots, 17}_{11 \text{ times}}, \quad \underbrace{289, \dots, 289}_{5 \text{ times}}.$$

Therefore

$$[R : \delta_G R + \delta_R R] = 17^{21} = 69\,091\,933\,913\,008\,732\,880\,827\,217.$$

Thus $(\delta_G, \delta_R)R[1/17] = R[1/17]$. Both determinants belong to $I_{1312}(E)$, although they use different row and column selections. Hence $I_{1312}(E)R_S = R_S$ by lemma 9.4. The structural kernel gives the upper bound 1312, so the rank is exactly 1312. $\square$

Remark 9.7 (Which primes are inverted). *The determinantal ideal already becomes the unit ideal over $R[1/17]$. The additional inversion of 2 and 3 is used when identifying component rank with the semisimple representation-theoretic obstruction, because $|\Gamma| = 2^2 \cdot 3 \cdot 17$.*

Remark 9.8 (Exact boundary). *Theorem 9.6 closes one of the six large component gates. The global gates for*

$$\rho\omega/\mathcal{B}_1, \quad \rho/\mathcal{B}_2, \quad \rho\omega/\mathcal{B}_2, \quad \rho/\mathcal{B}_3, \quad \rho\omega/\mathcal{B}_3$$

remain open. The theorem does not prove $\text{Supp}(\Theta_i) \subseteq \{2, 3, 17\}$, determine the global Smith groups, or make a tomography or Markov-connectivity claim.

10 Discussion, prior art, and open directions

10.1 The rational–integral contrast

The minimum regulus trades generate every rational relation in all three configurations. Integrally, we prove saturation at 2, 3, 17, 137, 219 097, 3 288 036 131. The global determinantal-ideal certificate closes one of the six large component gates, namely $\rho/\mathcal{B}_1$, away from 2, 3, 17. Equality of the full trade and relation lattices remains open because the other five large component gates have not been computed. Thus

$$\text{minimum local geometry} \implies \text{full rational generation} \not\Rightarrow \text{global integral saturation}.$$

The character table also shows that common incidence parameters and a common group do not make the relation modules isomorphic.

Cho’s characteristic-zero null-design decomposition and minimum subspace null-design work provide the complete-Grassmann background [2, 3]. Krotov supplies the minimum-support theorem and the $t = 1$ classification [8]. The ordinary-subset orbit-spanning result of Ghorbani–Kamali–Khosrovshahi is the closest structural analogue of theorem 5.2 [4]. Maliakas–Stergiopoulou develop total trades through intersection matrices and Specht modules [5]. Both ordinary-subset settings use the symmetric-group module structure; the present theorem concerns selected binary subspace constituents and the order-204 group.

For toric ideals, Aoki–Takemura–Yoshida own the indispensable-fiber criterion [1]. Betti–Grate–Holleben–Salizzoni develop incidence-toric/null-design language for ordinary subset incidence matrices [11]. The present finite calculation shows that the cubic fibers of these selected Grassmann configurations have exactly the two regulus monomials. Classical integral-design modules and circuit lattices are represented by [7, 6]. Determinantal ideals, cyclotomic arithmetic, CRT reconstruction, and Smith normal form are standard tools [12, 13, 14, 15]. These comparisons delimit scope; no method-firstness or priority claim is made.

10.2 Tomography and the P42 interface

The 97 155 three-dimensional subspaces of $\mathbb{F}_2^8$ split into 515 Γ -orbits, with orbit-size profile

$$51^5, \quad 102^{70}, \quad 204^{440}.$$

The selected three-space versus four-space incidence map

$$W_{3,4} : \mathbb{Q}[\mathcal{B}_i] \longrightarrow \mathbb{Q}[\text{Gr}_2(8, 3)]$$

has column rank 66 929. It is therefore injective on the entire block module and, in particular, on every K_i , giving a lossless rational sensor system for the relations.

This sensing statement is not used in theorems 5.2, 8.3 to 8.5 and 9.6; conversely, no tomography statement is used in the integral component certificates. Systematic comparison of sensor families, minimal sensor sets, and reconstruction belongs to the planned P42 tomography paper. The synchronized interface is

P39	P42
relation modules and circuit generators	measurement maps and reconstruction
one lossless $W_{3,4}$ check	systematic sensor comparison
integral component audit	sensing loss versus lifting loss.

10.3 Remaining global integral gates

At 137, 219 097, and 3 288 036 131, all six large maps have structural rank, giving complete three-design vanishing by theorems 8.3 to 8.5. Theorem 9.6 goes beyond that finite list for one map. The remaining integral tasks are:

- compute an exact localized order-maximal determinantal-ideal certificate for each of the other five large component maps;
- combine those results with proposition 7.6 to prove or refute $\text{Supp}(\Theta_i) \subseteq \{2, 3, 17\}$;
- compute the exact global Smith invariant factors, including the remaining 2-, 3-, and 17-primary structure.

A further finite-prime search cannot close these questions. The $\rho/\mathcal{B}_1$ result shows the required certificate shape: a common integral cyclotomic basis, exact characteristic-zero minors of the complete equation matrix, and their ideal sum. The five repetitions are deliberately deferred; none is needed for the rational, cubic, selected-prime, or one-component global theorems proved here.

10.4 Geometric and family questions

Krotov’s bound gives support 30 and leg volume 15 for $T_2(2, 4, 8)$, but the $t = 1$ uniqueness theorem used here does not classify that case. An exact next question is a two-colour census: one leg in $\mathcal{B}_i$, the other in $\mathcal{B}_j$. Exhaustiveness requires a separate classification theorem.

A second direction is to replace the locked finite triple by a general selected q -subspace configuration. The missing theorem is nonzero projection of the selected minimum trades to every null-design constituent. The component-rank criterion in lemma 5.1 isolates what such a theorem must prove.

Finally, higher-degree parts of I_i remain open. Cubic indispensability neither implies cubic generation nor determines the complete Markov or Graver basis. The P39 and P42 programs meet at

$$\mathbb{Q}\text{-generation} \longrightarrow \mathbb{Z}\text{-lifting} \longrightarrow \text{Markov connectivity},$$

but these stages should remain separate until their integral and higher-degree obstructions are classified.

A Exact certificate contracts

The abstract arguments in the paper reduce the locked finite statements to exact character, rank, determinant, and enumeration inputs. This appendix specifies what must be retained for those inputs to be independently checkable.

A.1 Characters and relation modules

The representation certificate contains:

1. the 24 conjugacy-class representatives and class sizes;
2. the action convention for a, b, c and a check of the group laws;
3. fixed-point counts on $\mathcal{P}, \mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$;

4. the absolute character table, Galois grouping, and rational type labels;
5. exact inner products giving the point, block, and kernel multiplicities;
6. dimension and class-function reconstruction checks.

Changing a class label, type ordering, or action convention must invalidate the reconstruction rather than silently permute a table.

A.2 Minimum-trade census

The census contract binds the digests and canonical orderings of the three block sets. Every accepted trade records Z, U , the two ordered reguli, the six lifted blocks, and the canonical unoriented sign convention. Verification checks:

1. all six blocks belong to the selected constituent;
2. the 255 point coordinates balance;
3. the restricted rank, five-column deletions, and active Smith form;
4. canonical deduplication and complete Γ -orbit closure;
5. agreement of template, membership, and orbit-counting passes.

Exhaustiveness is the combination of Krotov's $t = 1$ classification and complete enumeration of the corresponding (Z, U) -templates.

A.3 Rational component ranks

For each pair (i, V) , the full-kernel certificate stores:

1. block and point orbit representatives with stabilizers;
2. exact bases of V^H and the Frobenius-reciprocity coordinate convention;
3. the round-robin seed stream across all 69 two-space orbits;
4. the dimensions $m_{B_i}(V), m_P(V)$ and target rank;
5. row and column indices of a target-order square minor;
6. its nonzero determinant modulo 409;
7. an independent rank/minor reconstruction modulo 613;
8. a deletion or duplication control that loses the certified rank.

There are $3 \cdot 8 \cdot 2 = 48$ prime/component rank records. The 409 records are the characteristic-zero witnesses; the 613 records test construction stability.

A.4 Local saturation

At 2, the witness lists 66 674 explicit circuits per design. An independent replay reconstructs each point balance and obtains full rank with the opposite pivot convention.

At 3, the contract first verifies the Smith/Bockstein interface:

$$\text{rank } A_i = 254, \quad \text{rank}[A_i; \alpha] = 255,$$

together with the unique 3-divisible Smith factor 120. It then checks the F -representation laws, Wedderburn mass 68, all four component-family top ranks, and rank-loss mutations.

At 17, verification includes free C_{17} -actions, orbit counts 15 and 3 937, surjectivity on coinvariants, free kernel rank 3 922, the eight $C_4 \times C_3$ tops, Wedderburn mass 12, and one rank-loss control per component.

For the 18 small good-prime maps, the matrices and two selected maximal minors are reconstructed over the stated integral orders. Exact determinants and a Bezout gcd supported on $\{2, 3, 17\}$ certify the localized unit ideal.

A.5 The 137-component

The complete-equation contract for $\rho/\mathcal{B}_1$ retains:

1. the deterministic list of all 11 274 minimum-circuit orbit representatives in $\mathcal{B}_1$;
2. the dimension contract $1317 - 5 = 1312$;
3. a specified 1312-row, 1312-column submatrix of the complete equation matrix at $\zeta_{17} = 122$;
4. an independently reconstructed determinant $56 \bmod 137$;
5. duplicate-column, root-order, and input-hash controls.

For the remaining five large maps, the contract retains:

1. all primitive original columns and the entrywise exact-wide replay;
2. sixteen order-17 roots and, for $\rho\omega$, the sixteen Frobenius orbits in $\mathbb{F}_{137}[y]/(y^2 + y + 1)$;
3. eighty nonzero maximal-order determinant residues in total;
4. an independent conjugate-root/transposed reconstruction;
5. eighty duplicate-field-column determinant kills;
6. the combination with the eighteen small-component certificates.

Every localization contains a unit maximal-order minor, so no local-length calculation is required.

A.6 The 219 097-component

The complete-equation contract at $p = 219\,097$ retains:

1. a specified 1312-square submatrix of the complete $\rho/\mathcal{B}_1$ equation matrix at $\zeta_{17} = 91\,085$, with determinant $213\,269 \bmod 219\,097$;
2. target ranks and localization counts 16, 32, 16, 32, 16, 32 for the six large maps;

3. all determinant residues and root labels, with roots regenerated from the least primitive generator 5;
4. independent exact-wide parsers and transposed reconstructions;
5. all duplicate-column determinant kills, root-label mutation rejections, and input hashes;
6. the combination with the eighteen small-component exclusions.

Every complete localization contains a unit maximal-order minor.

A.7 The 3 288 036 131-component

The complete-equation contract at $p = 3\,288\,036\,131$ retains:

1. a specified 1312-square submatrix of the complete $\rho/\mathcal{B}_1$ equation matrix at root 1 120 906 769, with determinant 1 471 654 912 mod p ;
2. sixteen degree-one Φ_{17} primes and sixteen degree-two Φ_{51} primes in $\mathbb{F}_p[y]/(y^2 + y + 1)$;
3. target ranks and sixteen localizations for each of the six large maps;
4. all determinant residues, independent primitive-root enumeration, Frobenius-conjugate evaluations, transposed matrices, and duplicate-field- column controls;
5. scalar-root and quadratic-orbit mutation rejections and complete input hashes;
6. the combination with the eighteen small-component exclusions.

Every complete localization contains a unit maximal-order minor.

A.8 Global $\rho/\mathcal{B}_1$ structural rank

The characteristic-zero contract for theorem 9.6 retains:

1. the common $\mathbb{Z}[\zeta_{17}]$ exponent-basis lift of all 1317 columns and the complete circuit-equation universe;
2. the exact G-selection binary and manifest, with SHA-256

007D34CA9EBAB7268CA1CB55A1E81BF3E83B7983EE5FB8DD4300BEB41E13B155
B892C0A7BA9419A8C475FFACF05E63149E5A7F11AEBF443D0F01E96CCB0C96DF

and a direct determinant check at $(103, 8)$;

3. the independently selected reverse-row binary and manifest, with SHA-256

C3F857FDEAE82E8680B3EFB0AA281B59661E04E1E3E7C6D689A28A3FF7F45439
B2D696537A73C30E91D31A8FB3957CEF8F5826ACC8EED66E6881A38AB10EA82B

and a direct determinant check at $(137, 122)$;

4. the distinct row and column selection hashes for the two order-1312 submatrices of the same complete equation matrix;
5. every frozen split CRT prime and coefficient residue, the exact row-energy bounds, signed uniqueness inequalities, and explicit source-admitted status of the CRT generation transcript;
6. independent cyclotomic multiplication, all sixteen conjugates, exact norms, coefficient mutations, and rank-loss controls;
7. the Smith form of the 16×32 multiplication-lattice matrix, with diagonal $17^{11}, 289^5$ and index 17^{21} ;
8. byte-identical normal and optimized executions of the aggregate fail-closed verifier.

The two determinants need not be minors of one fixed 1312×1317 presentation. Their common ambient object is the complete equation matrix, and each therefore belongs to $I_{1312}(E)$. Their ideal sum becomes the unit ideal after 17 is inverted. This closes one component gate only; it does not certify the other five large maps or a complete Θ_i .

B Reproducibility and source boundary

The companion package is organized relative to its repository root. The principal fail-closed commands are

```
python scripts/check_manifest.py --closed-tree
python scripts/verify_all.py
python -O scripts/verify_all.py
python scripts/verify_manifest_only.py
```

The aggregate replay invokes four independent verifiers:

```
scripts/s449c_verify_g1r_replacement_minor.py
scripts/s449c_verify_g1r_principal_determinant.py
scripts/s449f_verify_reverse_row_determinant.py
scripts/s449f_verify_rho_d1_global_gate.py
```

The first and third parsers reconstruct their sparse binary matrices and directly check the displayed diagnostic determinants. The determinant reconstruction verifiers audit the frozen CRT prime lists and coefficient residues, prove signed uniqueness from row-energy bounds, recompute exact cyclotomic norms, construct both multiplication lattices, and reproduce the 16×32 Smith form. Generated JSON is canonical, and normal and optimized runs are required to be byte-identical.

Evidence tiers. The package contains the two exact sparse binary submatrices, their row and column selection manifests, the CRT inputs and outputs, and independent arithmetic verifiers for the global determinantal-ideal gate. The public scripts directly recompute the two diagnostic modular determinants. They do not regenerate all 189 CRT determinants from the binaries; those residues are source-admitted, hash-bound construction transcripts whose congruence, bound, uniqueness, norm, and mutation consequences are audited. The earlier census, rational-generation, local-saturation, and selected- prime theorem blocks are distributed as compact receipts. Their much larger raw matrix workspace remains in the source-locked P39 archive and is not duplicated in this reviewer-sized package. These boundaries are stated in the reviewer guide and claim ledger.

Evidence levels. Krotov’s classification and the indispensable-fiber criterion are imported theorems. The local circuit proof, semisimple criteria, Bockstein identification, Nakayama reductions, good-prime lemma, CRT uniqueness, multiplication-lattice interpretation, and determinantal-ideal implication are mathematical arguments in this paper. Character tables, exhaustive censuses, component ranks, exact minors, determinant coefficients, Smith diagonal, and covectors are exact finite inputs with independent replays or frozen hash-bound receipts. Resource stops from earlier wide Smith eliminations are not used as evidence.

Locked finite objects. The three block collections are identified by full canonical-key and point-mask SHA-256 hashes in `docs/SOURCE_LOCK.md`. Their construction route is pinned to the companion repository *FCIG-Common-Marked-Lattice-Designs* at commit `40b3e716...d7f3227`.

Source boundary. The prior-art search is dated and bounded. It found no source giving the same rational-spanning, cubic-fiber, or complete primary statements at 137, 219 097, and 3 288 036 131 for this locked triple, but that negative result is not a novelty or priority certificate. The paper makes no priority claim and does not generalize from the three selected configurations without a separate family argument. General determinantal-ideal, Smith, CRT, and cyclotomic tools are standard; no method-priority claim is made for their use in theorem 9.6.

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